



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Even Semester)

Innovative Teaching Method

UNIT I – Problem Solving

Degree, Semester & Branch: B.E., VI Semester & ECE A

Course Code & Title: C3491 & Artificial Intelligence and Machine Learning

Name of Faculty member: Ms.L.Krishna Kumari

Name of the Topic: Uninformed Search strategies

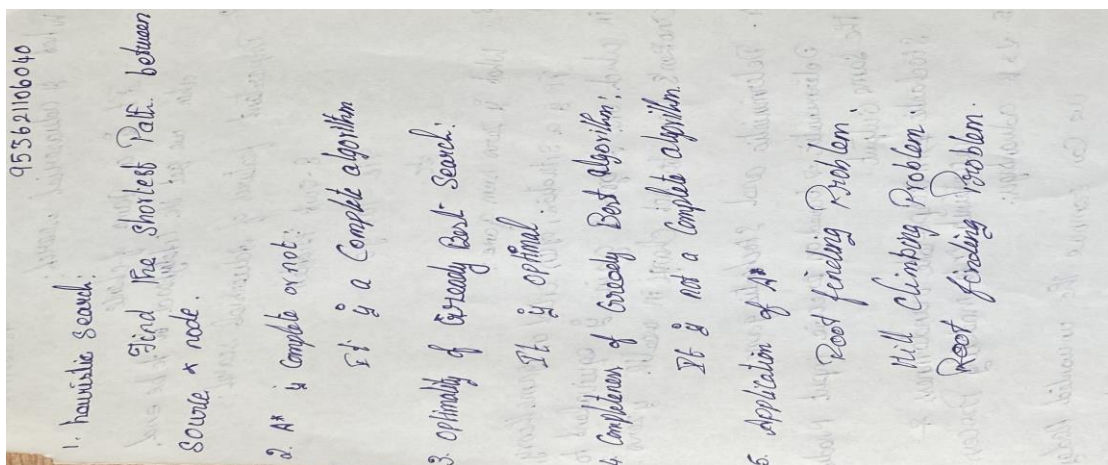
Name of the Innovative Practice: One-minute Paper

Date & Duration: 01.02.2024 & 5 minutes

Sample Images:

Innovative Teaching Method Execution

Uninformed Search strategies





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UNIT I – Problem Solving

Degree, Semester & Branch: B.E., VI Semester & ECE A

Course Code & Title: C3491 & Artificial Intelligence and Machine Learning

Name of Faculty member: Ms.L.Krishna Kumari

Name of the Topic: Adversarial Search

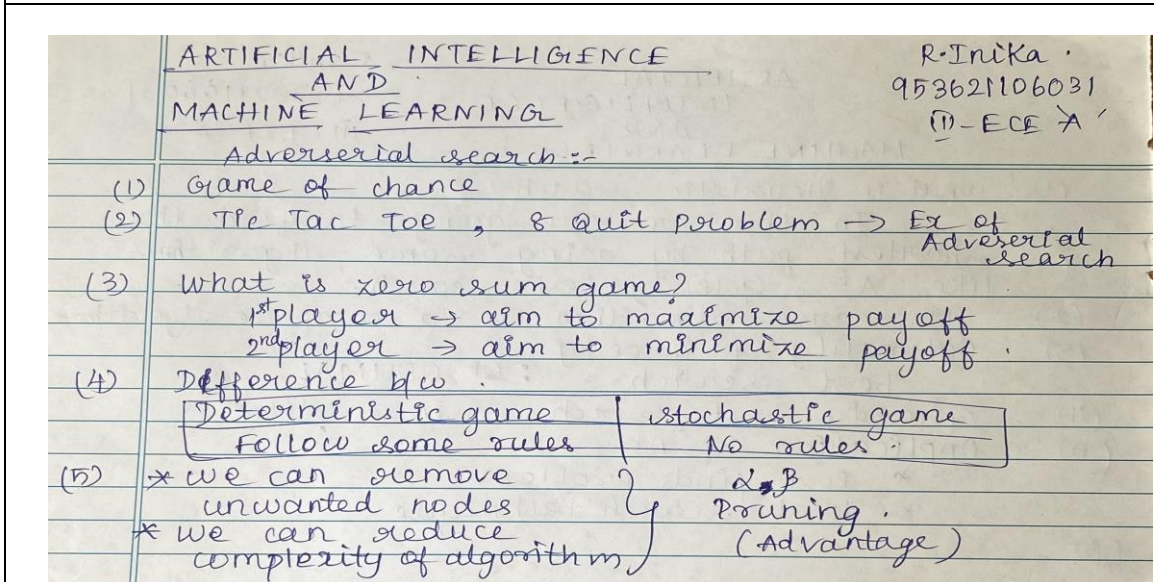
Name of the Innovative Practice: One-minute Paper

Date & Duration: 07.02.2024 & 5 minutes

Sample Images:

Innovative Teaching Method Execution

Adversarial Search





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Degree, Semester & Branch: B.E., VI Semester & ECE A

Course Code & Title: CS3491 & Artificial Intelligence and Machine Learning

Name of Faculty member (s): Ms.L.Krishna Kumari

Innovative Practice Description

- **Unit / Topic:** II/ Bayesian networks
- **Course Outcome:** CO2
- **Topic Learning Outcome:** 2b
- **Activity Chosen:** Jigsaw
- **Justification:**

The activity “JIGSAW” is used to recollect the knowledgeable concepts and to enrich the learning through this teamwork. A Bayesian network is a probabilistic graphical model which represents a set of variables and their conditional dependencies using a directed acyclic graph. Bayesian network and its inference strategies were taught to students through chalk and talk method. By recollecting all the information regarding Bayesian networks, students from expert group were able to share the information to home group.

- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

The students are divided into Home Groups with each group having 6 members. Each member in home group will form different groups named Expert group. Each expert group is allotted with each concepts in Bayesian networks. The expert group members discuss the topic in detail for 20 minutes with the learning materials and e-sources

The Expert group members should go to their original home group and discuss the points (25 Minutes) to the other members. All the members in home group learnt about all the topics through expert group members.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO11	PO12	PSO3
CO2	2	1	1	3	1	2	1	3	1

- **PO / PSO mapped:**

Innovative practice	PO1	PSO3
	2	1
Justification for correlation	The students could be able to use and apply basic probability concept to handle uncertainty situation. It is mapped at level 2.	The students could be able to apply the probability reasoning methods to develop machine learning model for solving the uncertainty problem. It is mapped at level 1.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

The students felt a new experience and they could able to recollect the information about various concepts of Bayesian networks and its real time applications. It helped them to learn the concept through collaborative learning with their class mates. They liked this practice of learning through this type of activity for recollecting the information that they already learnt.

- ❖ **Benefit of the practice:**

By forming the group, students were able to grasp the knowledge of concept by sharing the ideas among them. Each Home group could recollect the information of directed acyclic graph and its probability calculation.

❖ ***Challenges faced in implementation:***

Initially, I have planned the activity for 45 minutes but it takes some 5 minutes more time to share the information.

References:

Sl. No.	Title of Book	Author (s)	Publication with ISBN
1	Artificial Intelligence – A Modern Approach	Stuart Russell and Peter Norvig	Fourth Edition, Pearson Education, 2021.
2	Introduction to Machine Learning	Ethem Alpaydin	MIT Press, Fourth Edition, 2020
3	Introduction to Artificial Intelligence and Expert Systems	Dan W. Patterson	Pearson Education, 2007
4	Artificial Intelligence	Kevin Night, Elaine Rich, and Nair B	McGraw Hill, 2008



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Academic Year 2023 – 2024 (Even Semester)

Innovative Teaching Method

UNIT III – Supervised Learning

Degree, Semester & Branch: B.E., VI Semester & ECE A

Course Code & Title: C3491 & Artificial Intelligence and Machine Learning

Name of Faculty member: Ms.L.Krishna Kumari

Name of the Topic: Probabilistic generative model

Name of the Innovative Practice: One-minute Speech

Date & Duration: 20.03.2024 & 5 minutes

Sample Images:

Innovative Teaching Method Execution

Probabilistic generative model





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Engineering Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: B.E., VI Semester & ECE A

Course Code & Title: CS3491 & Artificial Intelligence and Machine Learning

Name of Faculty member : Ms.L.Krishna Kumari

Innovative Practice Description

- **Unit / Topic:** IV/ Expectation Maximization
- **Course Outcome:** CO4
- **Topic Learning Outcome:** 4c
- **Activity Chosen:** Flipped Class room
- **Justification:**

The topic chosen for flipped classroom is an expectation–maximization (EM) algorithm. Unit four deals with ensemble techniques and unsupervised learning. The expectation -Maximization algorithm can be used to handle situations where variables are partially observable. When certain variables are observable, we can use those instances to learn and estimate their values. Then, we can predict the values of these variables in instances when it is not observable This topic can be learnt in self-learning mode with simple manipulations and logic. The students have the chance to understand the performance various steps in EM algorithm. And find more information related to EM Algorithm in real time scenario. Thus this flipped class activity is chosen to make them to work out different steps in EM algorithm.

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

The overview of flipped class activity was given to students before the implementation of the activity. The students were divided into groups and one topic is allotted to each group from fiber optic measurement techniques as their wish. The topic related information and e-sources were given to the students. The students have to prepare a presentation for 10 minutes and each group has to present their topic chosen and others will listen to the presentation. After the presentation, question & answering is planned and others can share their ideas and doubts. By this way, all groups were shared their concepts learnt in their topic inside the class.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 2
CO4	3	1	3	1	1	2	1	2	1	1	1	1

CO4: Apply ensembling and unsupervised models for real world applications.

- **PO / PSO mapped**

Innovative practice	PO3	PSO3
	2	2
Justification for correlation	The students could be able to use the basic mathematical knowledge to build the combination of basic learning model and un supervised machine learning models. It is mapped at level 3.	The students could be able to develop mobile app with ensemble learning and un supervised ML model for any prediction or classification of real-world information and communication related problem. It is mapped at level 1.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

The students felt a new practice and it helped them to know to make good presentation and avoid stage fear. They liked this practice of teaching that can be done through this type of activity. They could get idea from learning sources which would help them to prepare the presentation.

❖ ***Benefit of the practice:***

Students done this activity in a successful manner. They planned the activity for their chosen topic and prepared the presentation in the conceptual manner. All members in the group were learnt the topics thoroughly. By forming the group, students were able

to grasp the knowledge of concept by sharing the ideas among them. Each member in the group was able to present and deliver the concept of optical fiber analysis and design.

❖ ***Challenges faced in implementation:***

Some of the students felt difficult to present because of lack of communication. Few groups done their presentation beyond their time limit.

Few students in the group felt difficult to answer all the questions.

References:

Sl. No.	Title of Book	Author (s)	Publication with ISBN
5	Artificial Intelligence – A Modern Approach	Stuart Russell and Peter Norvig	Fourth Edition, Pearson Education, 2021.
6	Introduction to Machine Learning	Ethem Alpaydin	MIT Press, Fourth Edition, 2020
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Academic Year 2023 – 2024 (Even Semester)

Innovative Practices Description

UNIT V – Neural Networks

Degree, Semester & Branch: VI Semester B.E.ECE A

Course Code & Title: CS3491 – Artificial Intelligence and Machine Learning

Name of the Faculty member: Ms.L.Krishna Kumari

Name of the Topic: Hyperparameter tuning

Name of the Innovative Practice: Think pair share

Date & Duration: 22.04.2024 & 30 Minutes

ICT Tool Used: Smart Class Room

Description:

Think-pair-share (TPS) is a collaborative learning strategy where students work together to solve a problem or answer a question about an assigned reading. This strategy requires students to think individually about a topic or answer to a question; and share ideas with classmates. Discussing with a partner maximizes participation, focuses attention and engages students in comprehending the reading material.

Goals (Learning Outcomes):

- The students will be able to Implement simple NN model for standard data set.

Use of appropriate methods:

Justification for choosing the topic using Think pair share activity:

Since the students are asked to think on the particular topic, they are able to think, remember and discuss the topics with their friends in the classroom. They are able to share their ideas on the particular topic by having discussions with them.

Effective presentation Implementation (Plan & Execution) with Proof:

The students were in the class room. The topic was explained to the students first. Then they were asked to think about the topic and after sometime they were asked to pair themselves and to have a discussion on the particular topic.

Innovative Teaching Method Execution

Think pair share on “Hyperparameter tuning”



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when the students were able to answer to the question on the particular topic when asked on the next week.
- ✓ The students understood the topic very clearly since they had discussion with their classmates.

Reflective Critique:

• Benefits:

The students were more interested in learning the concepts if they think and discuss the topic and ask questions. They clarified their doubts with the neighboring students and the faculty. They learnt the concepts of various techniques involved in hyper parameter tuning by think pair share activity. They asked more doubts in the topic and learnt many things.

• Challenges:

Basically some time is needed to explain the basic concepts of hyper parameter tuning. The slow learners found somewhat difficult in the initial stage later they also were able to think and share the concepts with their friends.

CO – PO / PSO mapping:

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 2
CO5	3	1	1	2	2	2	2	2	3	1	1	1

CO 5. Build deep learning NN Model

References:

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9	Artificial Intelligence – A Modern Approach	Stuart Russell and Peter Norvig	Fourth Edition, Pearson Education, 2021.
10	Introduction to Machine Learning	Ethem Alpaydin	MIT Press, Fourth Edition, 2020
11	Introduction to Artificial Intelligence and Expert Systems	Dan W. Patterson	Pearson Education, 2007
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